POLAR: Policy Optimization for Literature Analysis under Review Constraints

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Abstract

Systematic reviews are vital for evidence-based decision-making but remain resource-intensive due to the volume of literature requiring expert screening. Technology-Assisted Review (TAR) systems offer a solution by ranking documents for review, yet questions remain about how best to allocate limited human effort across multiple review topics. In this paper, we explore the problem of effort distribution by comparing alternative screening policies under fixed effort constraints. Using real-world data from the CLEF eHealth 2017–2019 TAR tasks, we evaluate both baseline and adaptive policies that account for topic size, screening depth, and residual uncertainty. We introduce effort-aware evaluation metrics to measure trade-offs between review effectiveness and resource use. Our results show that simple, topic-sensitive policies can significantly improve the yield of relevant documents discovered, offering practical insights for scalable and equitable systematic review workflows.

CCS Concepts

• **Information systems → Information systems applications**; **Retrieval models and ranking**; **Retrieval tasks and goals**; **Evaluation of retrieval results**.

Keywords

Technology-Assisted Review, Budget-Aware Evaluation, Effort Allocation Policies

ACM Reference Format:
Giorgio Maria Di Nunzio. 2025. POLAR: Policy Optimization for Literature Analysis under Review Constraints. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25), November 10–14, 2025, Seoul, Republic of Korea.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3746252.3760834

1 Introduction

Systematic reviews support evidence-based decision-making across domains such as medicine, public health, and policy. Unlike narrative literature reviews, they adopt structured protocols to exhaustively identify, assess, and synthesize relevant studies addressing a focused research question. This rigorous methodology ensures transparency and reproducibility but demands extensive human effort, often spanning months and consuming thousands of reviewer hours [13, 17].

CIKM '25, Seoul, Republic of Korea

ACM ISBN 979-8-4007-2040-6/2025/11
https://doi.org/10.1145/3746252.3760834

This burden is especially heavy in medical fields, where accurate and up-to-date reviews can directly influence clinical decisions and patient outcomes. [15]. The scale and velocity of biomedical literature, accelerated by open-access repositories and preprint archives, has rendered traditional manual review pipelines insufficient. To address this bottleneck, Technology-Assisted Review (TAR) systems have emerged as key enablers of scalable evidence synthesis. These systems, combining active learning, document ranking, and human-in-the-loop decision-making, aim to prioritize relevant studies and reduce reviewer workload [1, 7]. Despite their promise, TAR systems face a critical operational challenge: how to allocate limited human effort across multiple topics [2, 3, 8]. In practice, review projects often span diverse research questions or subdomains, each requiring a share of finite annotation budgets. Yet, most existing TAR frameworks are optimized at the level of single-topic document ranking, assuming idealized settings where resource constraints are absent or uniformly applied.

In this work, we argue that the deployment of TAR systems should be reframed as a problem of policy optimization under global effort constraints. Specifically, we focus on cross-topic screening policies: strategies that determine how annotation budgets should be distributed across multiple topics in order to maximize review utility. By moving beyond isolated topic optimization, we highlight a largely unexplored design space with substantial implications for real-world evidence synthesis. Our empirical analysis leverages the CLEF eHealth TAR benchmarks from 2017 to 2019 [9, 10, 14], which simulate realistic screening conditions. We reinterpret these experiments through the lens of policy evaluation, and introduce a new suite of budget-aware evaluation metrics that quantify the trade-offs between document yield and resource consumption. Our contributions are:

- Policy-Driven Re-Evaluation: We reinterpret CLEF TAR tasks as constrained screening problems and introduce a framework for comparing cross-topic budget allocation policies.
- Effort-Aware Metrics: We propose two novel metrics, Relevant Found per Cost Unit and Utility Gain at Budget, that capture screening efficiency under limited budgets.
- Strategy Benchmarking: We systematically evaluate baseline policies including uniform, proportional, and greedy strategies under varying topic distributions.

2 Policy-Aware Evaluation of TAR Under Budget Constraints

Traditional evaluations of TAR systems emphasize ranking quality within individual topics, typically using metrics such as Recall@k or Precision@k [1, 12, 18]. While these metrics are useful for assessing early retrieval effectiveness, they fail to capture the practical

Table 1: Definitions of the four allocation policies for topic t_i with total budget B.

Even	Proportional	Inverse Proportional	Greedy (capped)
$\pi_{\text{even}}(t_i) = \frac{B}{\lvert T \rvert}$	$\pi_{\text{prop}}(t_i) = \frac{D_i}{\sum_j D_j} \cdot B$	$\pi_{\text{inv}}(t_i) = \frac{1/D_i}{\sum_j (1/D_j)} \cdot B$	$\pi_{\text{greedy}}(t_i) = \min(\tau \cdot D_i, B_{\text{remaining}})$

constraints and multi-task nature of real-world systematic reviews. Reviewers often face a global annotation budget, reflecting constraints in time, financial resources, or cognitive effort, which must be allocated across multiple screening topics. This setting introduces a higher-order challenge: determining how to allocate effort across topics. We propose to model this challenge as a constrained policy optimization problem where the goal is to design a policy π that maximizes utility across topics under a global budget B.

2.1 Problem Formulation

Let $T = \{t_1, ..., t_n\}$ denote a set of n topics to review, each associated with a ranked list of documents produced by a TAR system. Let B be the total annotation budget, i.e., the total number of documents that reviewers can screen across all topics. A budget allocation policy π defines how to distribute this budget across the topics:

$$\pi : T \to \mathbb{N}, \quad \text{such that} \sum_{i=1}^{n} B_i = B$$

where $B_i = \pi(t_i)$ is the number of documents reviewed in topic t_i.

Each document review incurs a cost (e.g., time or money), and each relevant document yields a utility gain. We denote by: c_i the unit cost of reviewing a document in topic t_i; g_i the gain of identifying a relevant document in topic t_i; $y_{ij} \in \{0, 1\}$ the binary relevance of the j-th ranked document in t_i; $\mathcal{U}(\pi)$ the total utility obtained under policy π which is equal to:

$$\mathcal{U}(\pi) = \sum_{i=1}^{n} \sum_{j=1}^{B_i} u_{ij}, \quad \text{where} \quad u_{ij} = \begin{cases} +g_i & \text{if } y_{ij} = 1 \\ -c_i & \text{otherwise} \end{cases}$$

The objective is to find a policy π^* that maximizes total utility under a global budget:

$$\pi^* = \arg\max_{\pi} \mathcal{U}(\pi) \quad \text{s.t.} \quad \sum_{i=1}^{n} B_i = B$$

This abstraction captures the essential trade-off in cross-topic TAR: allocating effort to uncertain, high-utility areas while avoiding waste on low-yield or low-priority topics. In this framework, each baseline allocation strategy can be interpreted as an instance of a simple policy class, enabling consistent comparison and evaluation.

2.2 Policy Classes for Budget Allocation

In this section, we reinterpret the baseline strategies as policies within a constrained decision-making framework. Each policy represents a different inductive bias about topic utility, size, and uncertainty. In Table 1, we summarize the definitions of these policies.

Even Allocation (π_{even}). A uniform policy assumes that all topics have equal utility potential and screening cost, and allocates equal budget to each topic.

Proportional Allocation (π_{prop}). A size-weighted policy assumes that larger topics are more informative or deserve more scrutiny, and allocates a budget proportional to the number of documents D_i in each topic.

Inverse-Proportional Allocation (π_{inv}). A compensatory policy aims for balanced topic coverage and is useful when relevance is sparse, therefore favoring smaller topics.

Capped-Greedy Threshold Allocation (π_{greedy}). A depth-first, size-capped policy that allocates budget sequentially based on a fraction $\tau \in (0, 1]$ of D_i or the remaining budget, $B_{\text{remaining}}$. Topics with smaller documents to screen are visited first. This policy prioritizes fully screening smaller topics before allocating effort to larger ones, thereby promoting broad topic coverage and improving early recall when resources are scarce.

3 Budget-Aware Policy Evaluation

In this section, we define evaluation metrics that quantify the efficiency and utility of an allocation policy π under realistic resource constraints. When operating under a fixed screening budget, the performance of a TAR system depends not only on the underlying ranking model but also on the policy that distributes review effort across topics. We formalize evaluation as a policy utility estimation problem, where a policy π distributes effort across a set of topics T, and receives feedback in the form of retrieved relevant documents.

3.1 Evaluation Metrics

Unlike traditional metrics, such as Recall, which assume either known recall targets or unconstrained annotation budgets, our proposed metrics operate under more realistic assumptions: 1) the total number of relevant documents is unknown at screening time; 2) reviewers operate under a fixed budget B; and 3) each document screening action has a cost, while identifying a relevant document yields measurable utility.

To illustrate these new metrics, we first consider the widely used Recall@k metric, which is defined for a given topic t_i as:

$$\text{Recall@}k = \frac{\text{TP}_k}{P_{\text{total}}}$$

where TP_k is the number of relevant documents retrieved in the top-k screened items, and P_{total} is the (usually unknown) total number of relevant documents for t_i. While Recall@k measures effectiveness, it ignores the cost of obtaining that recall and assumes access to complete ground truth, making it less suitable for cost-sensitive, real-world TAR evaluations.

Relevant Found per Cost Unit (RFCU@k). To capture the efficiency of document screening under cost constraints, we define the RFCU@k metric as:

$$\text{RFCU@}k = \frac{\text{TP}_k}{k \cdot c_i}$$

where: TP_k is the number of relevant documents found within the top k reviewed documents in topic t_i; c_i is the cost per document for topic t_i (e.g., time or effort). RFCU@k quantifies how efficiently the screening process converts review effort into relevant findings. It is especially valuable when document cost varies across topics (e.g., due to length or complexity) or when operational efficiency is critical. Unlike Precision@k (which is bounded in $[0, 1]$), RFCU@k is unbounded and interprets retrieval success relative to cost, not purity. For $c_i = 1$, RFCU@k reduces to Precision@k.

Utility Gain at Budget (UG@B). We further define a utility-centric metric that explicitly rewards relevant findings and penalizes wasted effort on non-relevant documents. For a given topic t_i with budget $B_i = \pi(t_i)$, we define:

$$\text{UG@}B_i = \sum_{j=1}^{n} u_{ij} \quad \text{where} \quad u_{ij} = \begin{cases} +g_i, & \text{if } y_{ij} = 1 \\ -c_i, & \text{if } y_{ij} = 0 \end{cases}$$

This function captures the net gain from executing policy π under realistic assumptions: utility increases with correctly retrieved relevant documents; utility decreases with effort spent on non-relevant documents. Parameters g_i and c_i can be tuned to reflect domain-specific trade-offs. For example, in time-critical reviews, $g_i \gg c_i$, whereas in highly resource-sensitive settings (e.g., low-resource countries), both values may be tightly constrained.

A key advantage of RFCU@k and UG@B is that they do not rely on full knowledge of the relevance distribution. This aligns with real-world TAR deployments. By focusing on observed utility within a bounded review budget, these metrics allow consistent and reproducible evaluation without oracle knowledge. This makes them appropriate for both retrospective analysis (e.g., CLEF benchmark runs) and prospective deployment.

3.2 Policy Evaluation Protocol

To compare alternative policies π, we apply each strategy across the ranked document lists of each topic in a benchmark dataset (e.g., CLEF TAR 2017–2019), using the same global budget B. For each run:

(1) The policy determines a topic-level budget allocation B_i.
(2) Each system's top-B_i documents for topic t_i are evaluated.
(3) Recall@k, RFCU@k and UG@B are computed per topic.

This policy-aware evaluation framework supports both absolute and comparative analyses of review efficiency. In Section 4, we use this framework to compare baseline policies and highlight implications for real-world systematic review planning.

To complement the evaluation, we introduce a baseline policy, $\pi_{epsilon}$, based on the adaptive ϵ-greedy multi-armed bandit framework [16]. This policy begins by allocating a small initial budget to each topic to obtain preliminary observations. During subsequent allocations, with probability $1 - \epsilon$ the policy exploits by assigning the next budget unit to the topic with the highest observed average reward so far, while with probability ϵ it explores by sampling a random topic. This simple adaptive strategy balances exploration of uncertain topics with exploitation of topics that have demonstrated high yield, and provides a stronger comparison point against fixed heuristic allocations under constrained budgets.

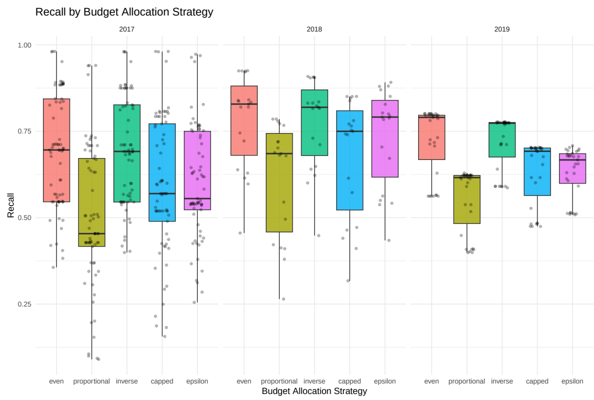
4 Experimental Results

We present empirical results evaluating alternative budget allocation policies applied to the CLEF eHealth Technology-Assisted Review (TAR) benchmark datasets from 2017 to 2019 [9, 10, 14]. Our primary goal is 1) to assess how different cross-topic effort allocation strategies perform under fixed annotation budgets, using the policy-aware framework introduced in Section 2 and 2) compare them with the adaptive $\pi_{epsilon}$ approach (with $\epsilon = 0.3$). We use the official participant runs from CLEF TAR tasks between 2017 and 2019.[1] These runs contain ranked document lists for multiple review topics derived from Cochrane Systematic Reviews, along with binary relevance annotations. Let T denote the set of topics in each year's task. For each run, we simulate the effect of different allocation policies π by redistributing a fixed global screening budget B across topics, reviewing the top $B_i = \pi(t_i)$ documents per topic. This section reports results from applying the four fixed allocation strategies and the adaptive ϵ-greedy approach to the CLEF official runs. All evaluations are performed under uniform cost and gain assumptions ($c_i = g_i = 1$) and a budget B_i equal to 10% of the total corpus size, approximating realistic resource constraints for systematic reviews. For the $\pi_{epsilon}$ approach, we used UG@B as the observed reward at each iteration.

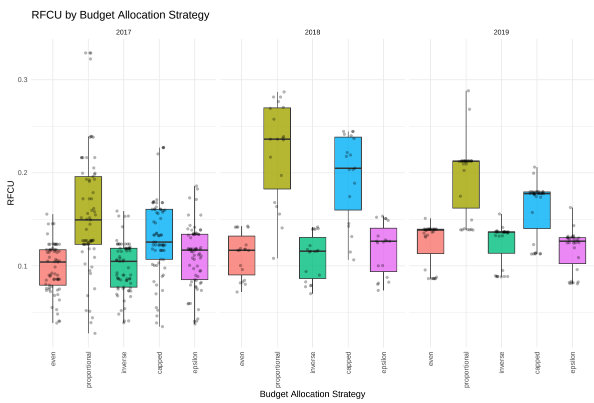
Policy Sensitivity of Evaluation Metrics. We evaluate each policy π according to three metrics introduced in Section 3. Figure 1a, Figure 1b, and Figure 1c display boxplots of each metric across all runs and topics, grouped by policy. Each point represents the average score of a run for that policy in that year. When optimizing for Recall@k, the best-performing strategies were *even allocation* and *inverse proportional.* These policies promote broad topic coverage, preventing over-allocation to large, low-yield topics. In particular, *inverse proportional allocation* excels at uncovering relevant items in small or underrepresented topics, improving topic-wise completeness. In contrast, the highest RFCU@k scores, measuring efficiency of relevance retrieval per cost unit, were achieved by the *proportional* and *threshold-capped greedy* strategies. These policies focus resources toward larger or early high-yield topics, and restrict overcommitment where returns diminish. This behavior aligns with RFCU emphasis on cost effectiveness rather than broad recall. Unexpectedly, UG@B scores exhibited minimal variance across all four strategies. No statistically significant differences were observed year-over-year. This suggests that the UG@B metric, by balancing gains and costs into a net utility value, is robust to variations in how budget is distributed, at least under the assumption of equal gain and cost. It may be more sensitive in future experiments with heterogeneous cost/gain ratios. The adaptive policy performs comparably to the less effective policies in terms of Recall and RFCU, but achieves the best performance for UG@B. This suggests, though it requires confirmation through further experiments, that the adaptive strategy may be significantly influenced by the chosen reward.

Metric Correlation and Trade-offs. To better understand the interplay between evaluation criteria, we analyzed the Pearson correlation between Recall@k, RFCU@B, and UG@B across CLEF TAR tasks from 2017 to 2019 - for space reasons, tables and figures are

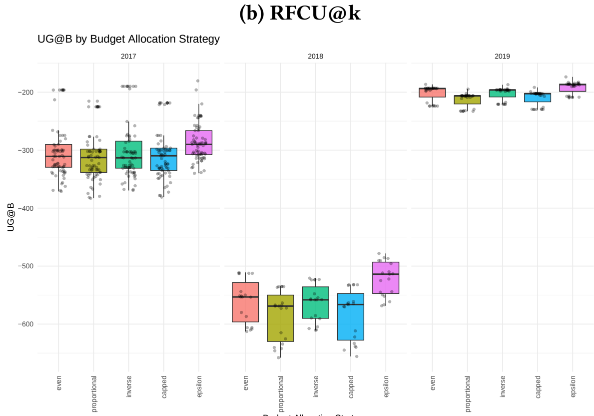
[1] https://github.com/CLEF-TAR



(a) Recall@k



(b) RFCU@k



(c) UG@B

Figure 1: Comparison of allocation policies across CLEF TAR tasks (2017–2019). Subplots report results using (a) Recall@k, (b) RFCU@k, and (c) UG@B metrics.

not shown here [6]. Our analysis reveals that RFCU@B and Recall@k are weakly correlated. This indicates that high recall does not necessarily imply high efficiency. In many cases, systems or allocation strategies that achieve broad Recall do so at a substantial annotation cost, resulting in low RFCU scores. Conversely, cost-efficient strategies may retrieve fewer relevant documents overall but do so with significantly less wasted effort. In contrast, UG@B exhibits a more moderate and consistent correlation with Recall@B. This reflects UG role as a balanced metric that rewards the retrieval of relevant documents while also penalizing unnecessary review of non-relevant items. It partially aligns with traditional Recall goals while incorporating cost sensitivity. Importantly, the strength and direction of these correlations vary across CLEF editions. These variations are influenced by differences in topic distributions, relevance prevalence, and TAR system behavior across years. This reinforces the importance of using multiple, complementary metrics when evaluating TAR systems, particularly under constrained budgets, to obtain a fuller picture of both effectiveness and efficiency trade-offs.

5 Conclusions

This paper examined the role of budget allocation strategies in the evaluation of TAR systems under realistic resource constraints. Through a retrospective analysis of the CLEF eHealth TAR benchmarks from 2017 to 2019, we demonstrated that allocation policies have a substantial impact on review outcomes. Even simple heuristic strategies can lead to meaningful differences in both effectiveness and efficiency, depending on the evaluation metric used. While stopping criteria and recall estimation remain important complementary challenges [4, 5, 11], our focus is on fixed-budget allocation strategies, assuming externally defined effort bounds. This perspective brings TAR evaluation closer to the constraints and trade-offs faced in real-world deployments. Our findings show that evaluation in budget-constrained settings is inherently multi-objective. No single policy consistently dominates across all metrics. Strategies such as even and inverse proportional allocation perform best when the goal is topic-wide coverage and high recall, while proportional and threshold-capped greedy approaches yield stronger results when cost efficiency is prioritized. The utility-based metric UG@B provides a more balanced view by incorporating both benefits and penalties, and appears less sensitive to the specific allocation policy. Lastly, the policies examined in this study are fixed and hand-crafted, the formal framework we present also supports the development of adaptive or learnable strategies. These may include uncertainty-aware allocation, where budget is dynamically adjusted based on prediction entropy or topic variance; bandit-style approaches, which model topics as arms with unknown reward distributions and optimize exploration-exploitation trade-offs. Although we leave the implementation of these strategies to future work, our framework provides a reproducible benchmark for their evaluation and comparison. To support transparency and reproducibility, the source code, experimental configurations, and evaluation scripts used in this study are available online.[2]

Acknowledgments

This work was partially supported by the "National Biodiversity Future Center - NBFC" project funded under the National Recovery and Resilience Plan (NRRP), Mission 4 Component 2 Investment 1.4 - Call for tender No. 3138 of 16 December 2021, rectified by Decree n. 3175 of 18 December 2021 of Italian Ministry of University and Research funded by the European Union – NextGenerationEU. Project code CN_00000033.

[2] https://github.com/gmdn/CIKM2025.git

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GenAI Usage Disclosure

The authors used OpenAI's ChatGPT to assist in proofreading and improving the clarity of the manuscript. The tool was employed exclusively for language refinement, including grammar corrections, sentence restructuring, and stylistic polishing. All technical content, experiments, analysis, and conclusions were developed solely by the authors. The use of generative AI did not contribute to the creation of any novel ideas, code, data analysis, or experimental design.